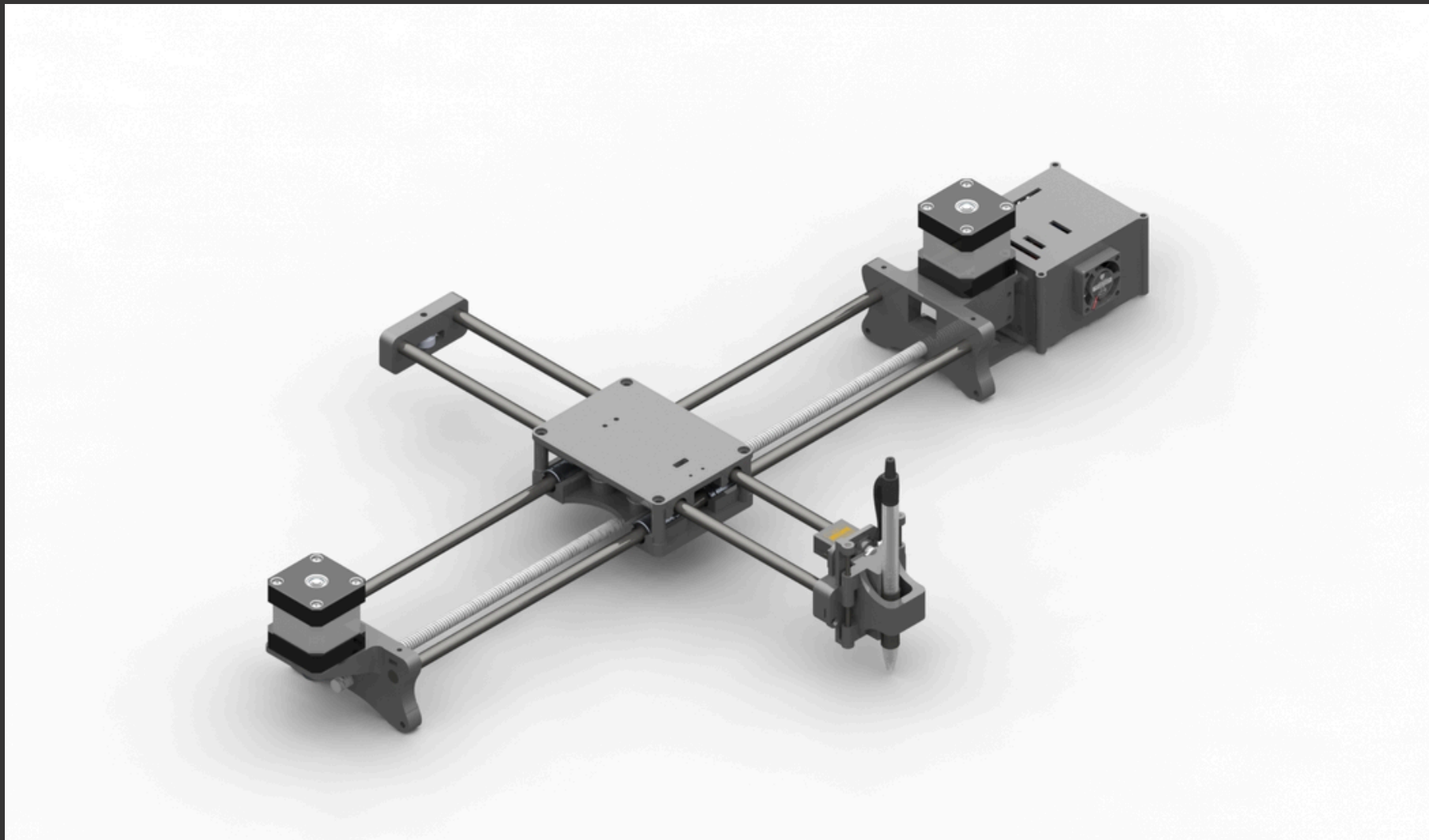


DRAWBOT



Kinematics & Dynamics Project

Members

Paing Thet Ko (65011416)

Rachata Raktham (65011500)

Sornpawee Jukmongkul (65011546)

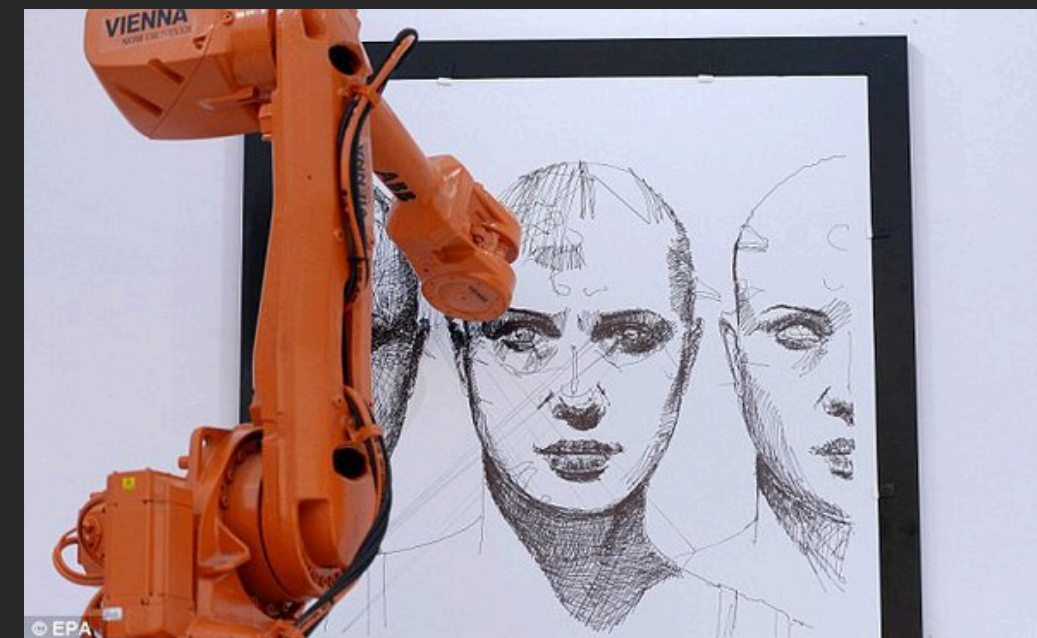
Swan Pyae Sone (65011683)

Thura Zaw (65011685)

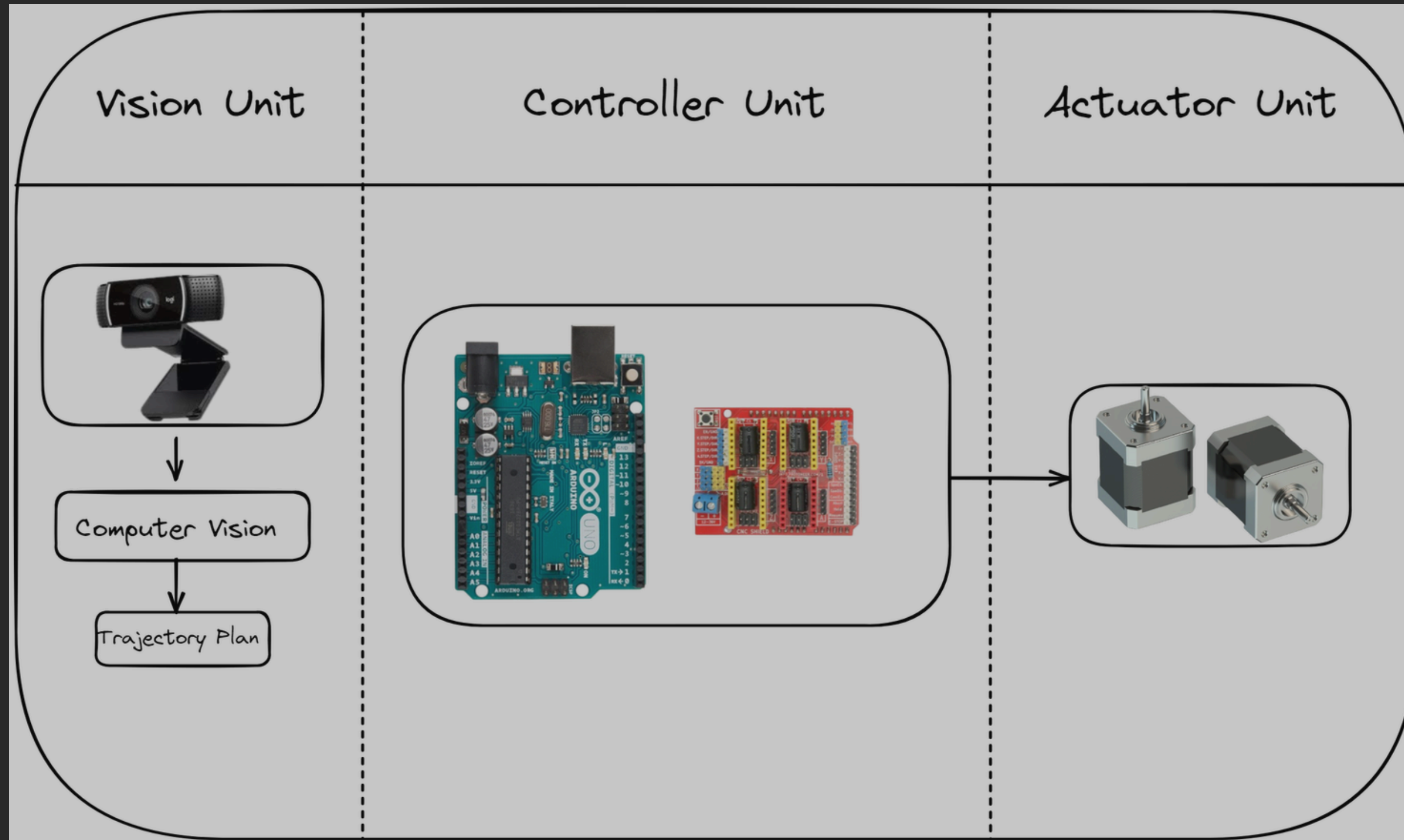
PURPOSE AND MOTIVATION

- Arts are once thought to be only created by human.
- The rise of computer vision and robot technology
- Applied Kinematics and Robot Control System
- Speed and Efficiency
Accessibility
- Budget Friendly

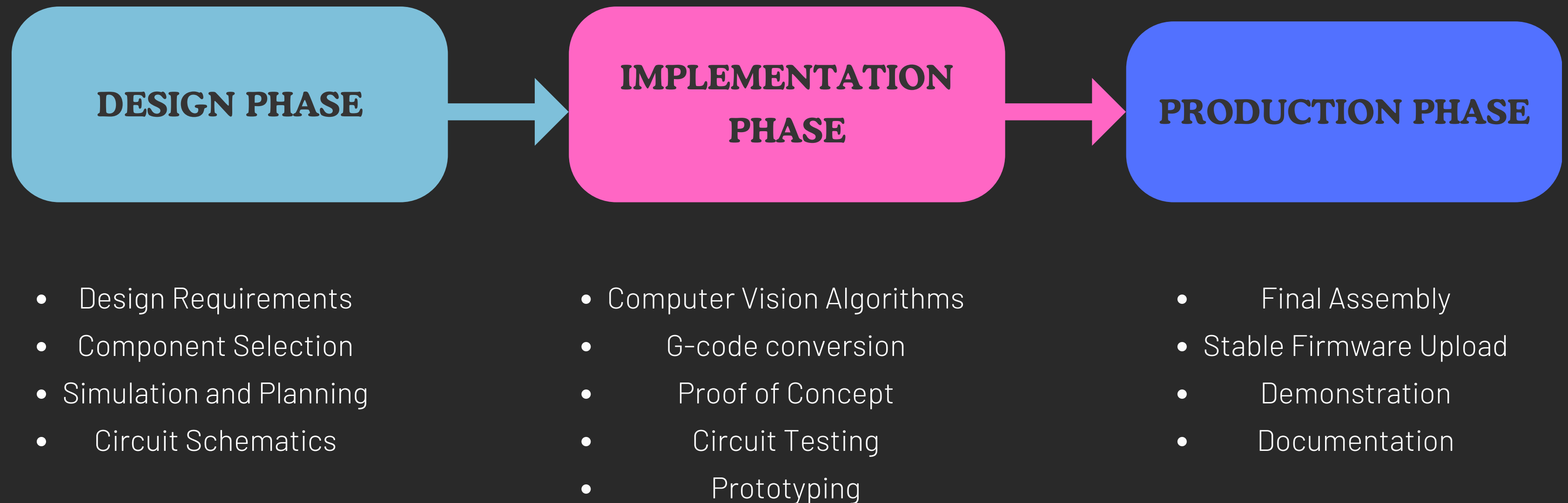
Understand how kinematics works

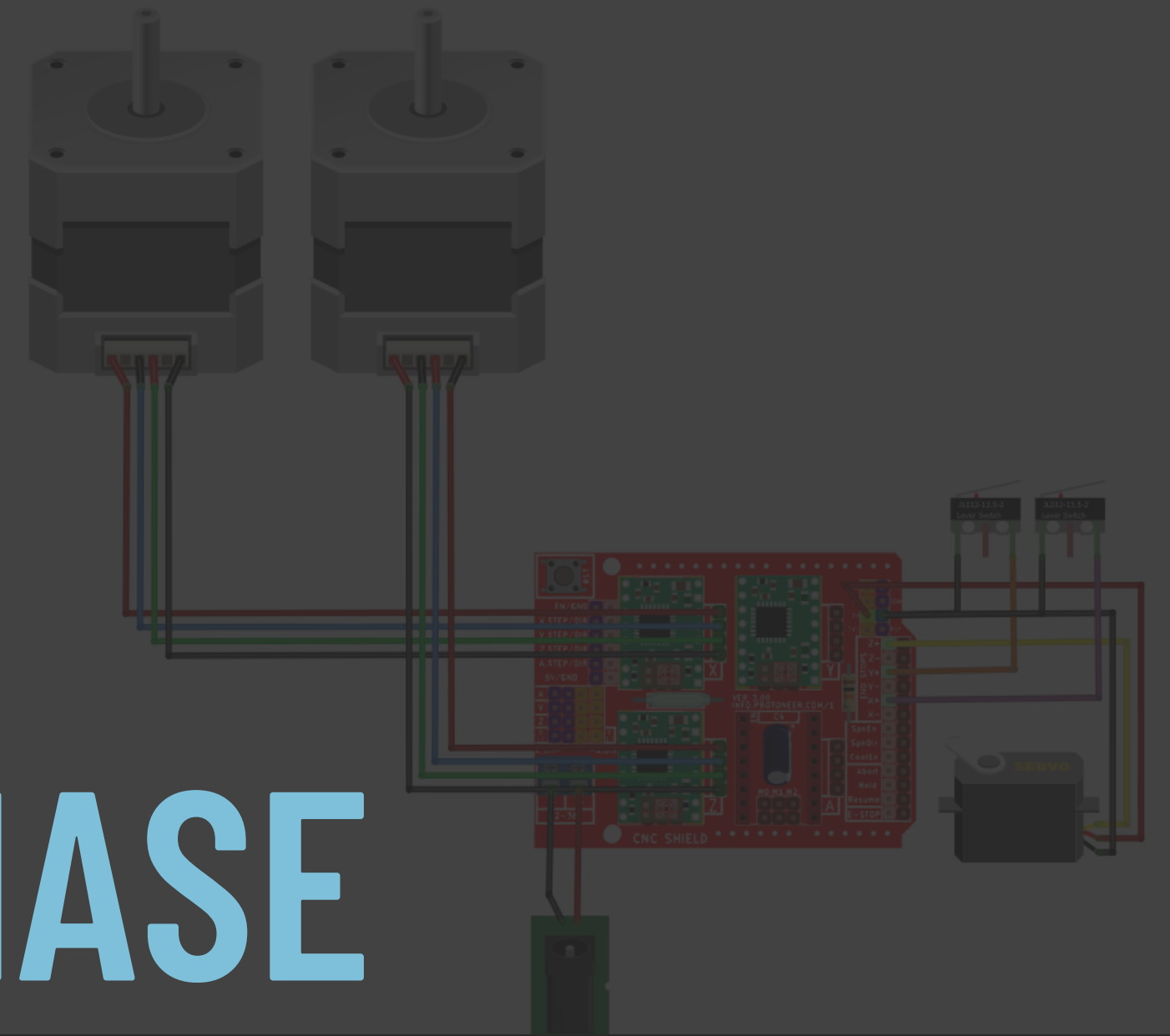


METHODOLOGY

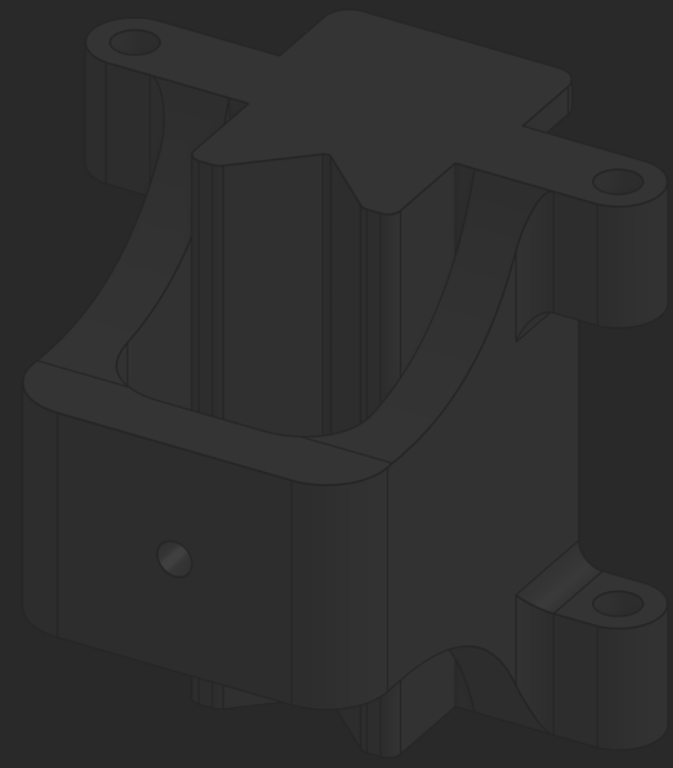


DEFINE GOALS





DESIGN PHASE



DESIGN REQUIREMENTS

- **Workspace**

- Rectangular (X & Y)
- Suitable Scheme (Cartesian)

- **Kinematics**

- Modified CoreXY Kinematics
- Parallel Link (Cable Driven Train) used in 3D printers

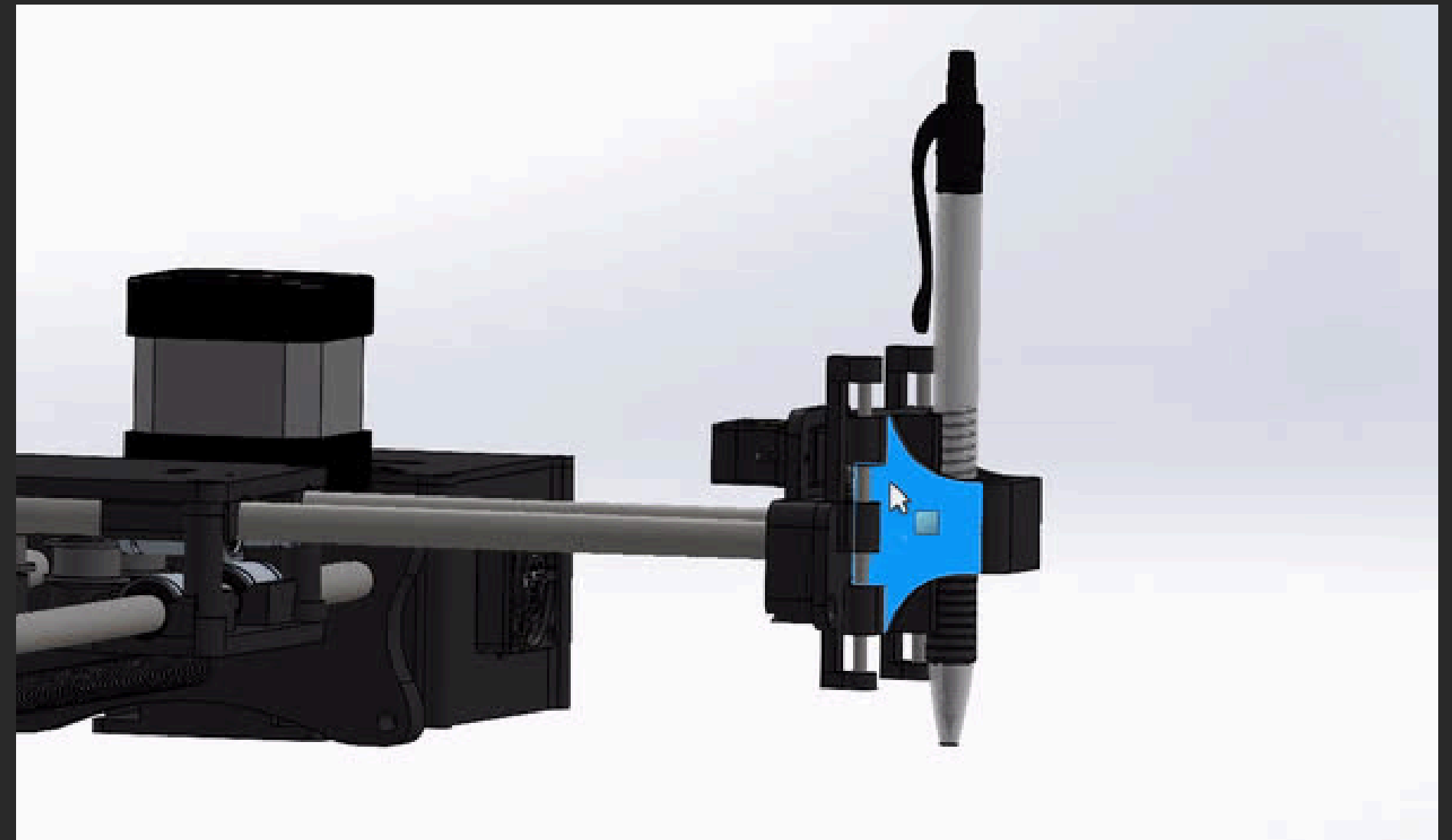
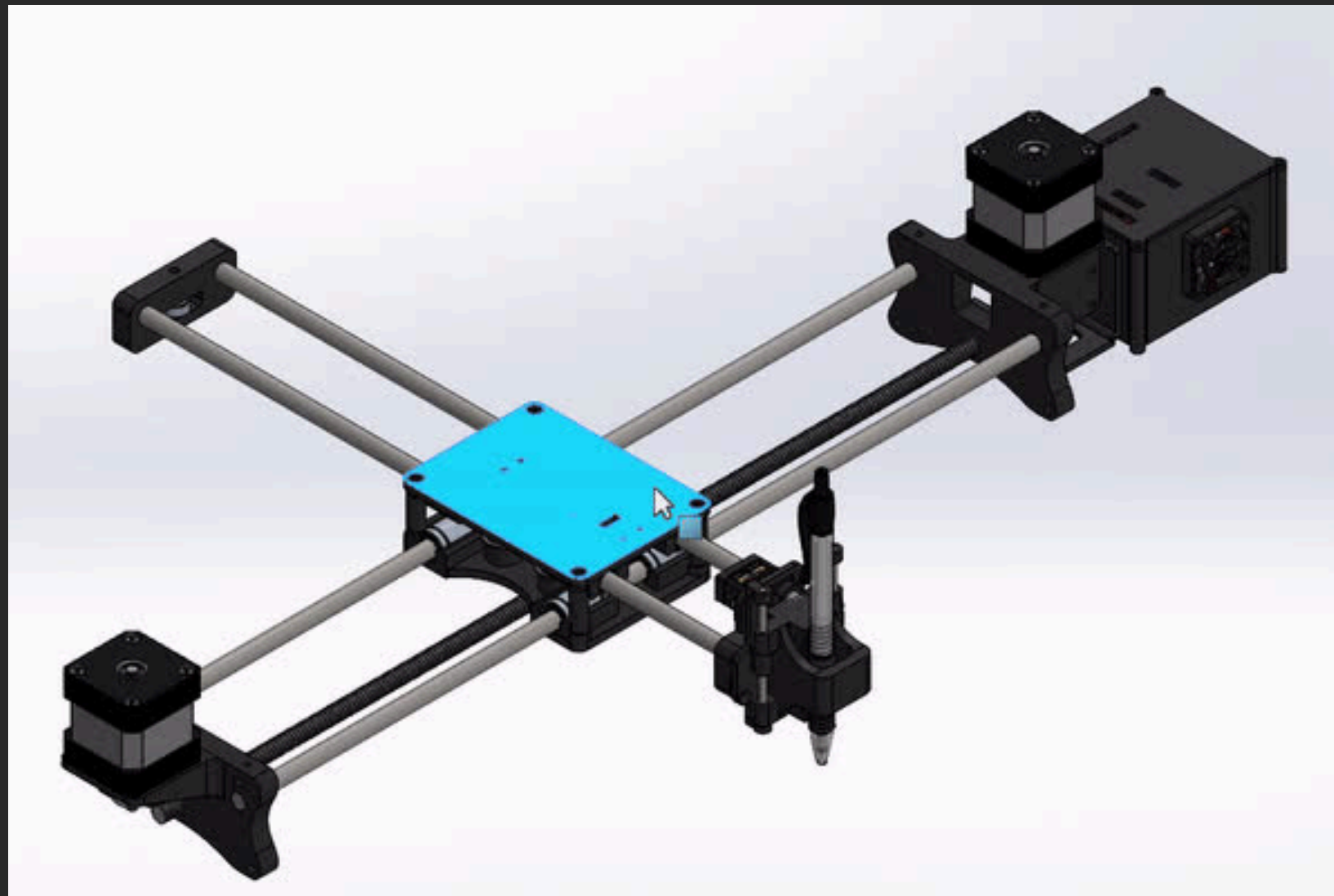
- **Actuators**

- Stepper motor (Precise positional control)
- 12 V AC-DC power supply (power hungry)
- Servo motor (Z-axis movement)

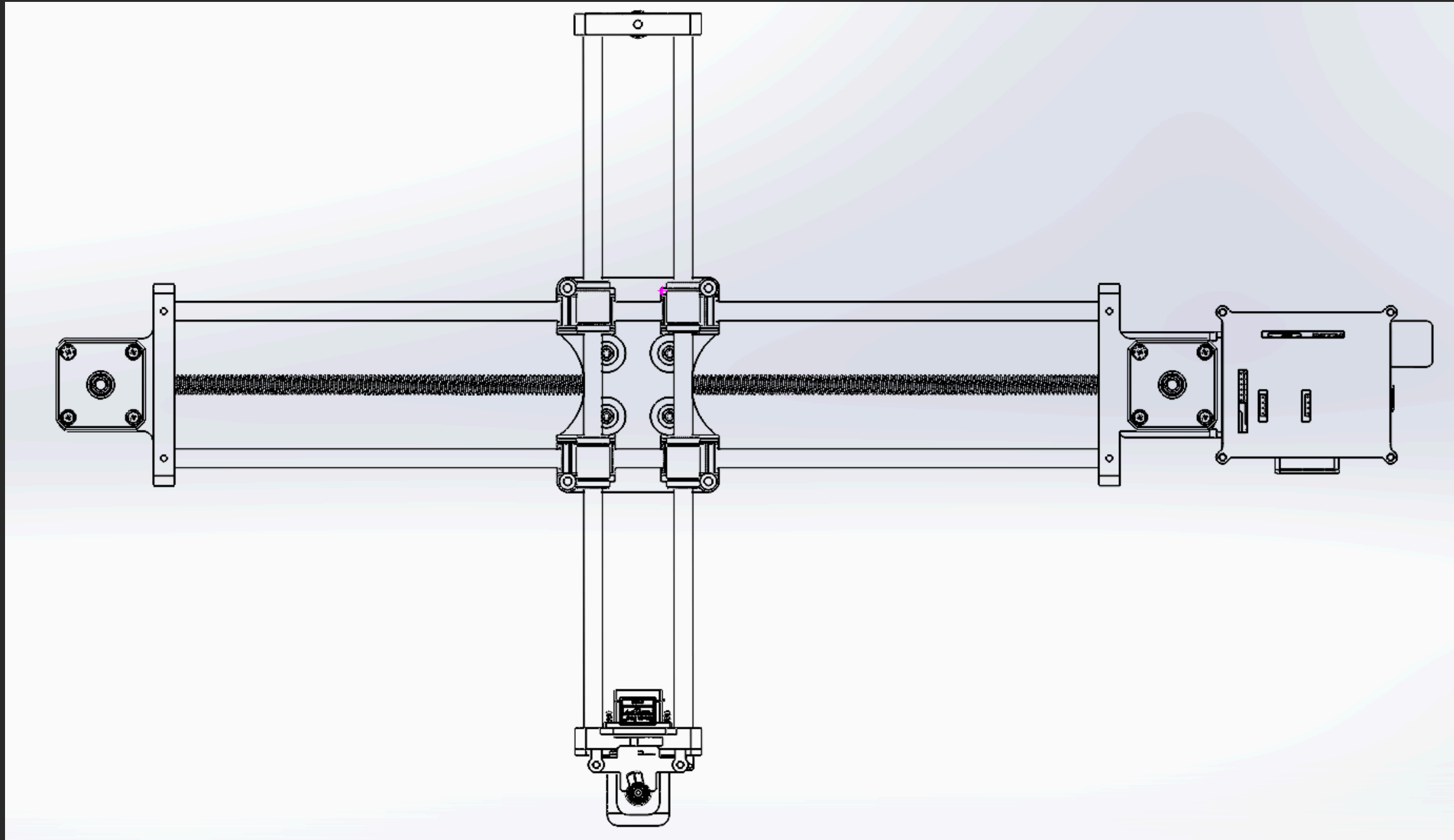
- **Controller**

- Arduino UNO (Sufficient EEPROM)
- Can control multiple motors with CNC Shield
- Can process GRBL firmware

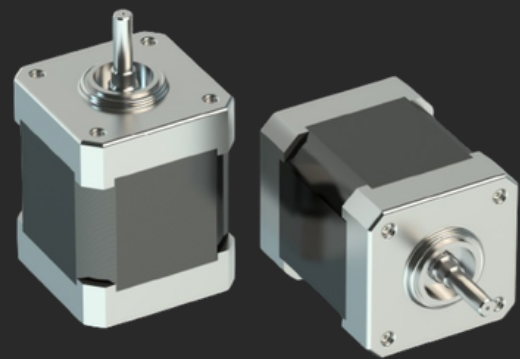
COMPUTER AIDED DESIGN (CAD)



COMPUTER AIDED DESIGN (CAD)



COMPONENT SELECTION



Stepper motors



Linear Rods



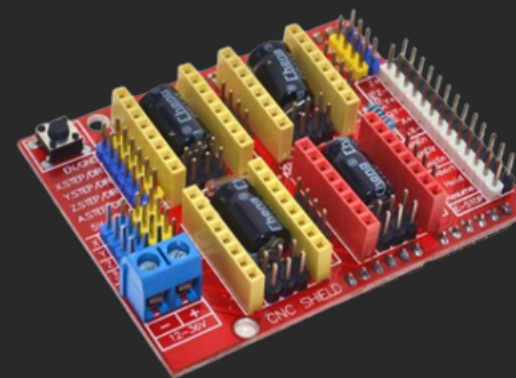
Bearings (Linear and Rotary)



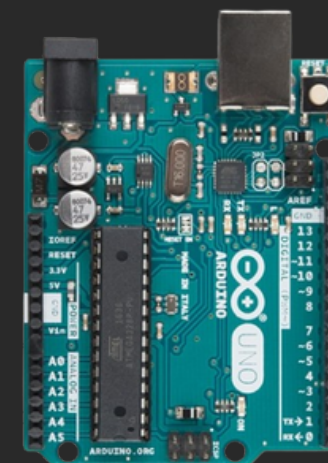
A bunch of screws and nuts



A4988 Stepper Driver



CNC Shield



Arduino UNO



Cooling Fan

KINEMATICS MODEL

$$x = \frac{1}{2}r(-q_1 - q_2)$$

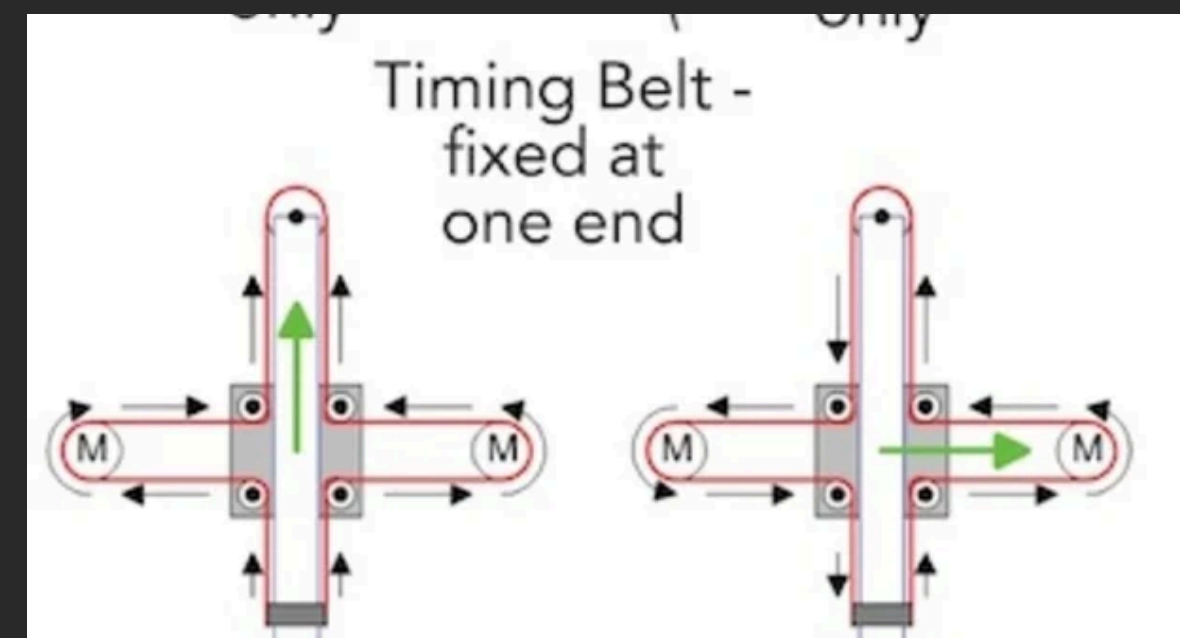
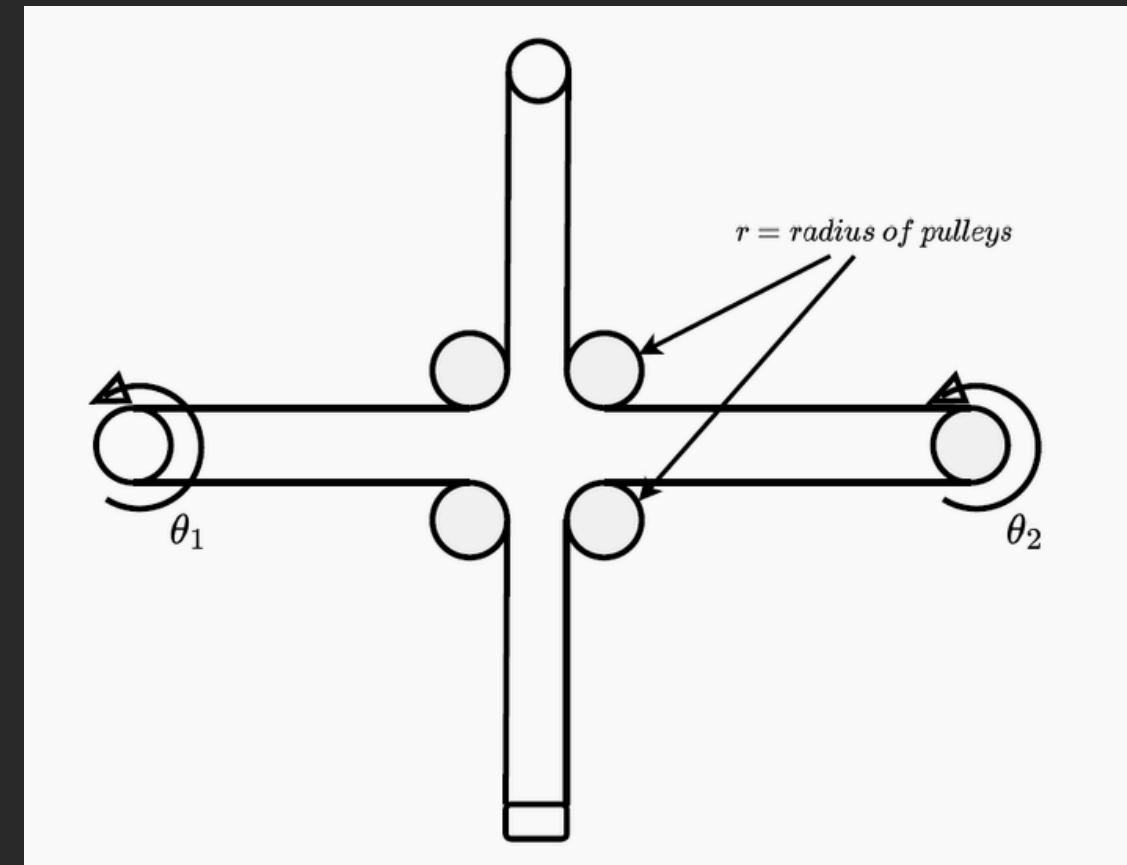
$$y = \frac{1}{2}r(q_1 - q_2)$$

Forward Kinematics

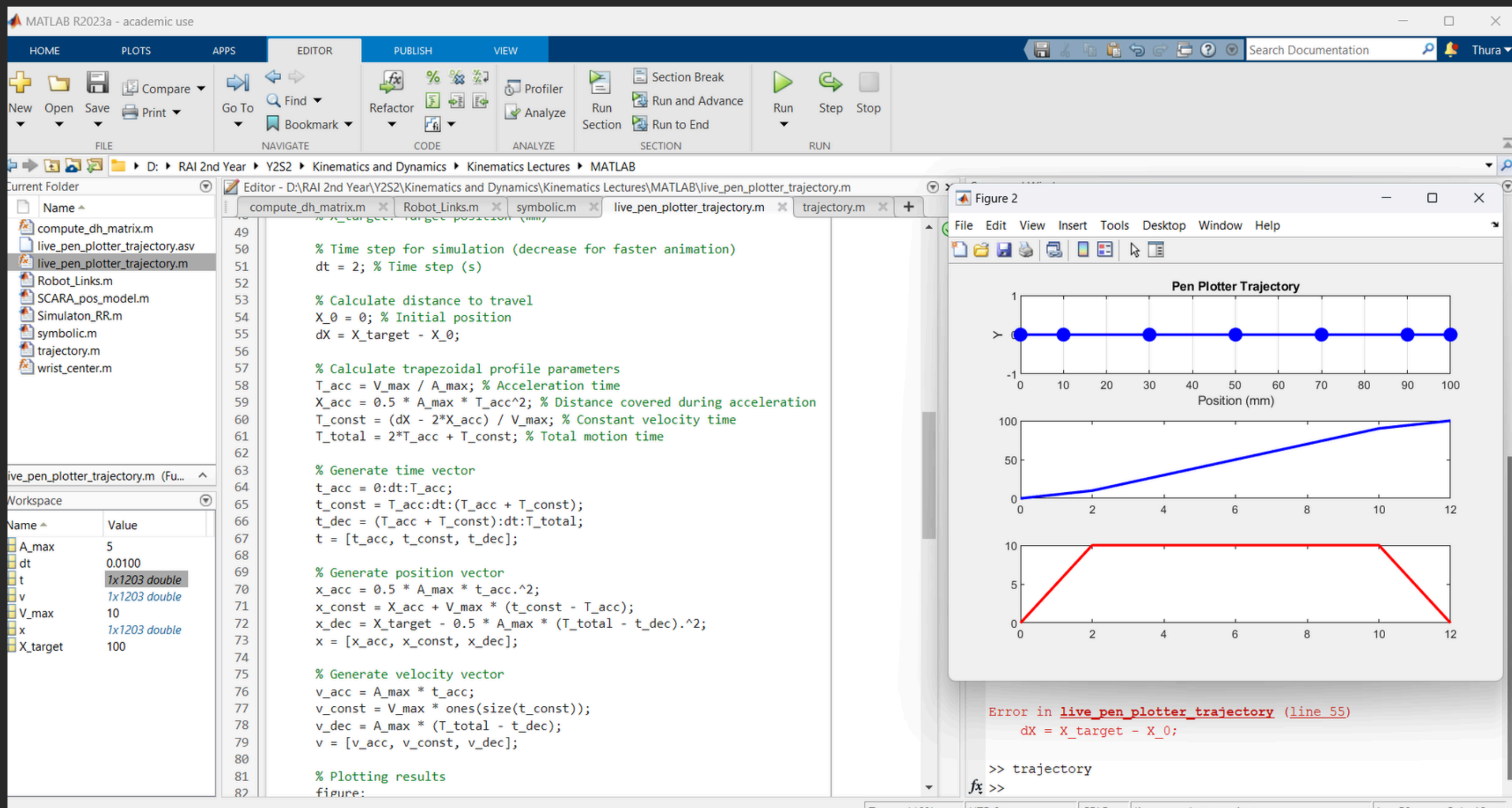
$$q_1 = \frac{1}{r}(-x + y)$$

$$q_2 = \frac{1}{r}(-x - y)$$

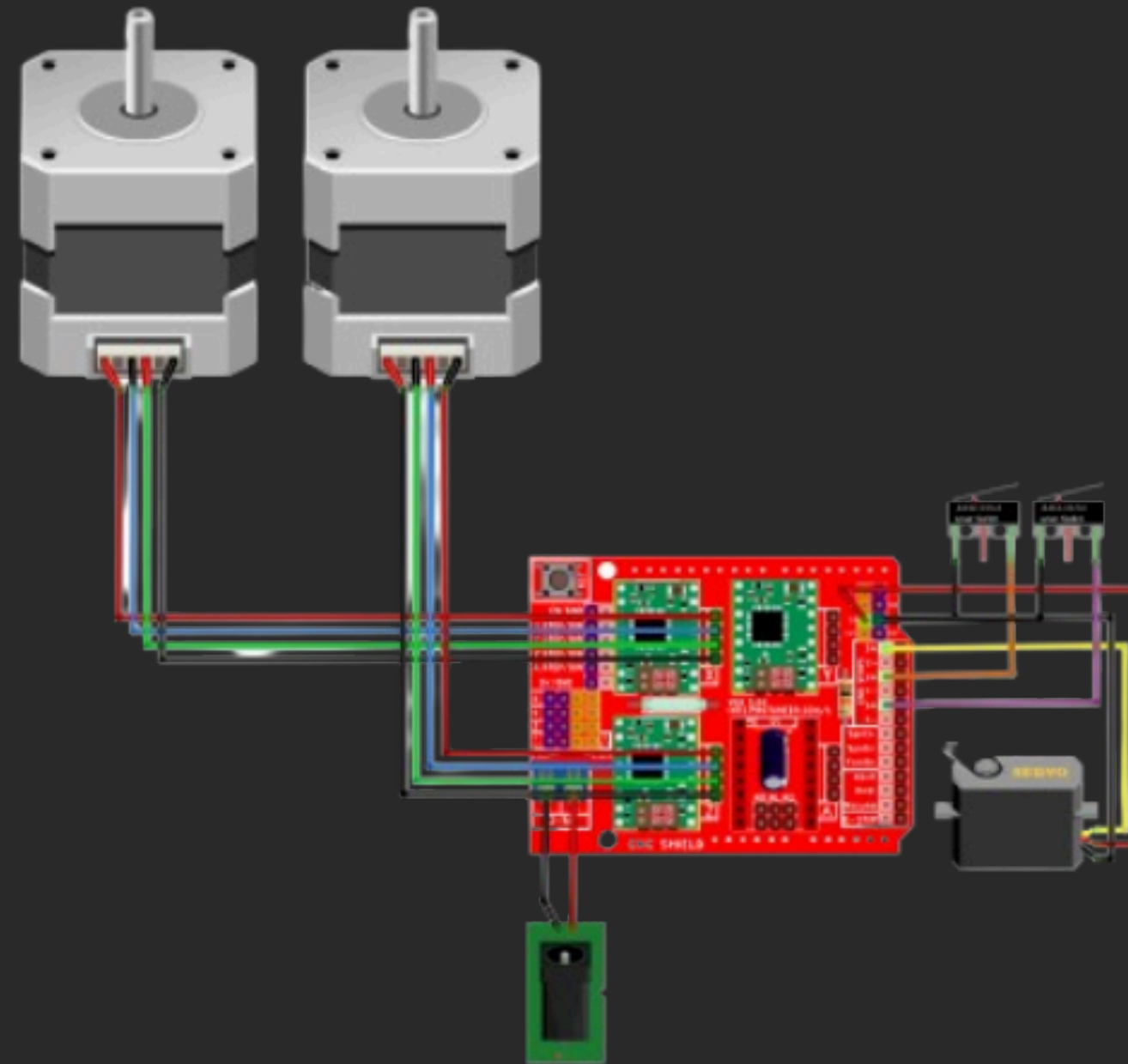
Inverse Kinematics

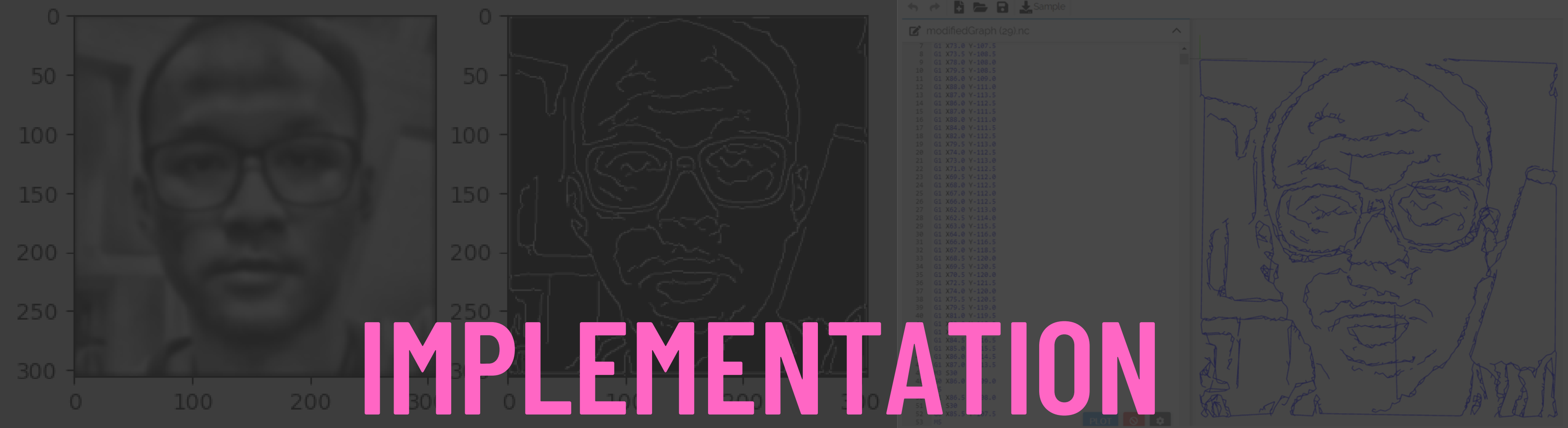


MATLAB SIMULATION OF TRAJECTORY PLAN

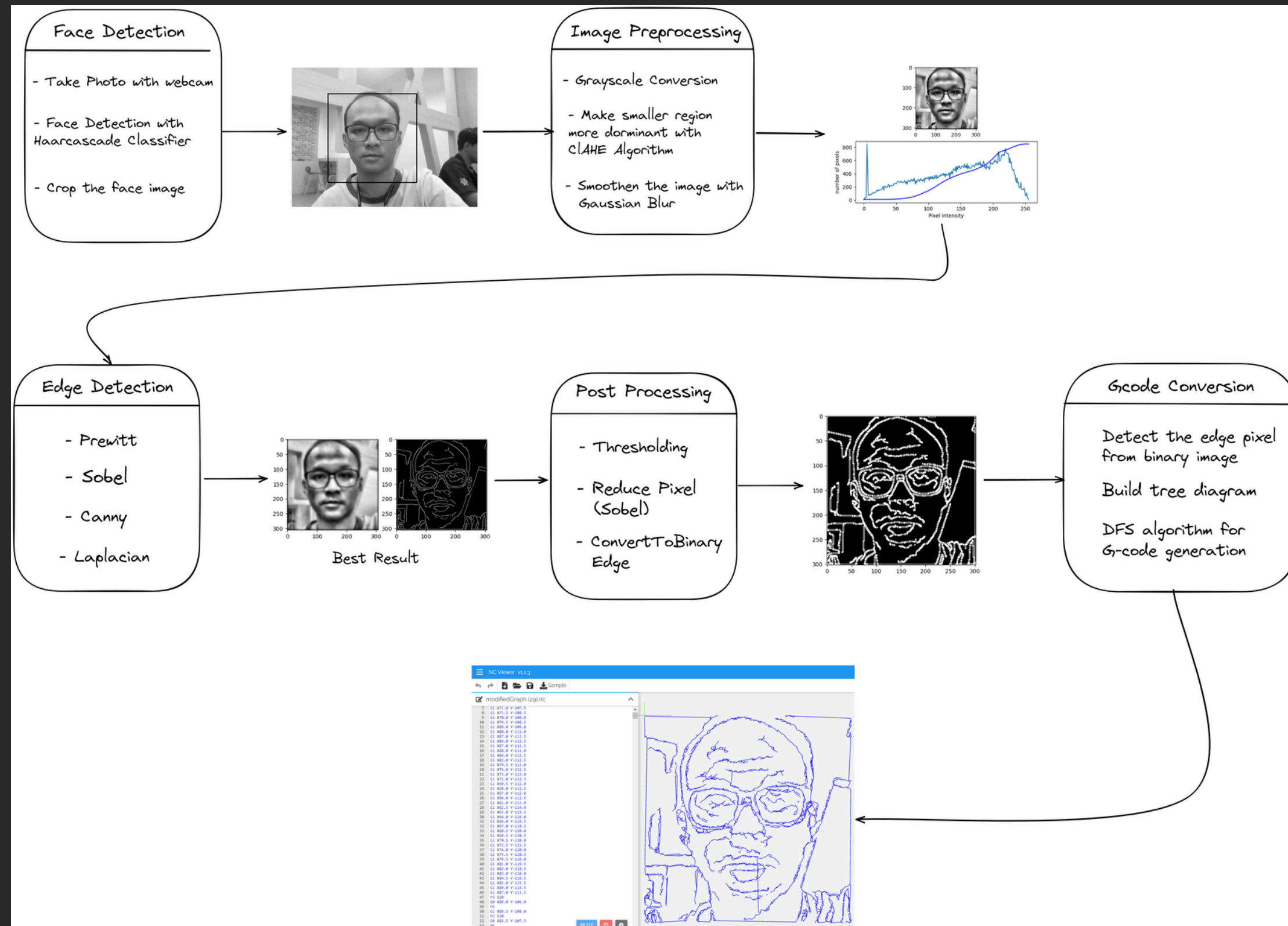


CIRCUIT SCHEMATICS





COMPUTER VISION ALGORITHMS



G-CODE REPRESENTATION

```
G1 X39.7502 Y43.8305
G1 X40.4883 Y44.4776
G1 X41.2263 Y45.1248
G1 X40.3386 Y45.7583
G3 X39.4037 Y46.3063 I-3.512 J-4.9205
G3 X35.8882 Y47.8557 I-32.3884 J-68.7261
G2 X30.6334 Y50.6062 I12.2165 J29.7343
G2 X27.8367 Y53.0688 I6.8968 J10.652
G3 X27.2988 Y53.4321 I-0.912 J-0.7705
G3 X26.1882 Y53.6128 I-1.1247 J-3.4079
G2 X22.9734 Y54.6784 I0.0213 J5.4467
G2 X21.3224 Y57.2069 I2.5727 J3.483
G2 X21.2524 Y61.4782 I10.192 J2.3033
G2 X22.8158 Y64.994 I8.3254 J-1.5966
G3 X23.7348 Y66.2987 I-18.8947 J14.2852
G3 X24.0917 Y66.9416 I-4.2052 J2.7549
G2 X25.4304 Y68.2777 I2.4839 J-1.15
G2 X28.4109 Y68.9312 I2.9805 J-6.4705
G2 X30.6843 Y68.6178 I-0. J-8.4042
G2 X34.833 Y67.1267 I-8.3901 J-29.8607
G3 X37.4439 Y66.0399 I26.3054 J59.5165
G3 X39.1974 Y65.4203 I10.2467 J26.208
G2 X40.9643 Y64.8346 I-20.8285 J-65.7912
G2 X42.1139 Y64.4097 I-9.7966 J-28.2709
```

G-CODE = GEOMETRY CODE (USED TO COMMAND THE TRAJECTORY OF THE ROBOT)

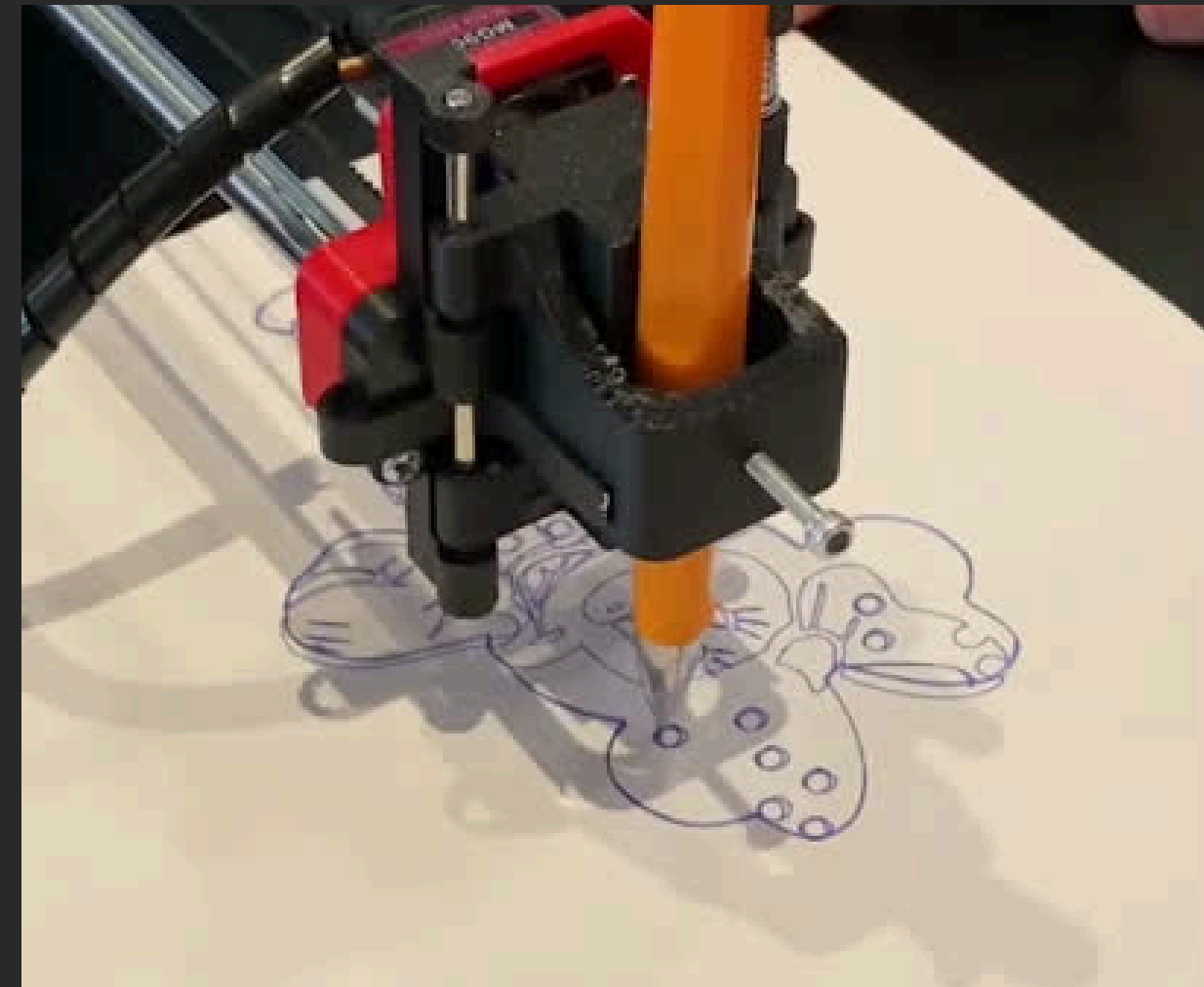
G0 - Rapid Movement from one contour to another

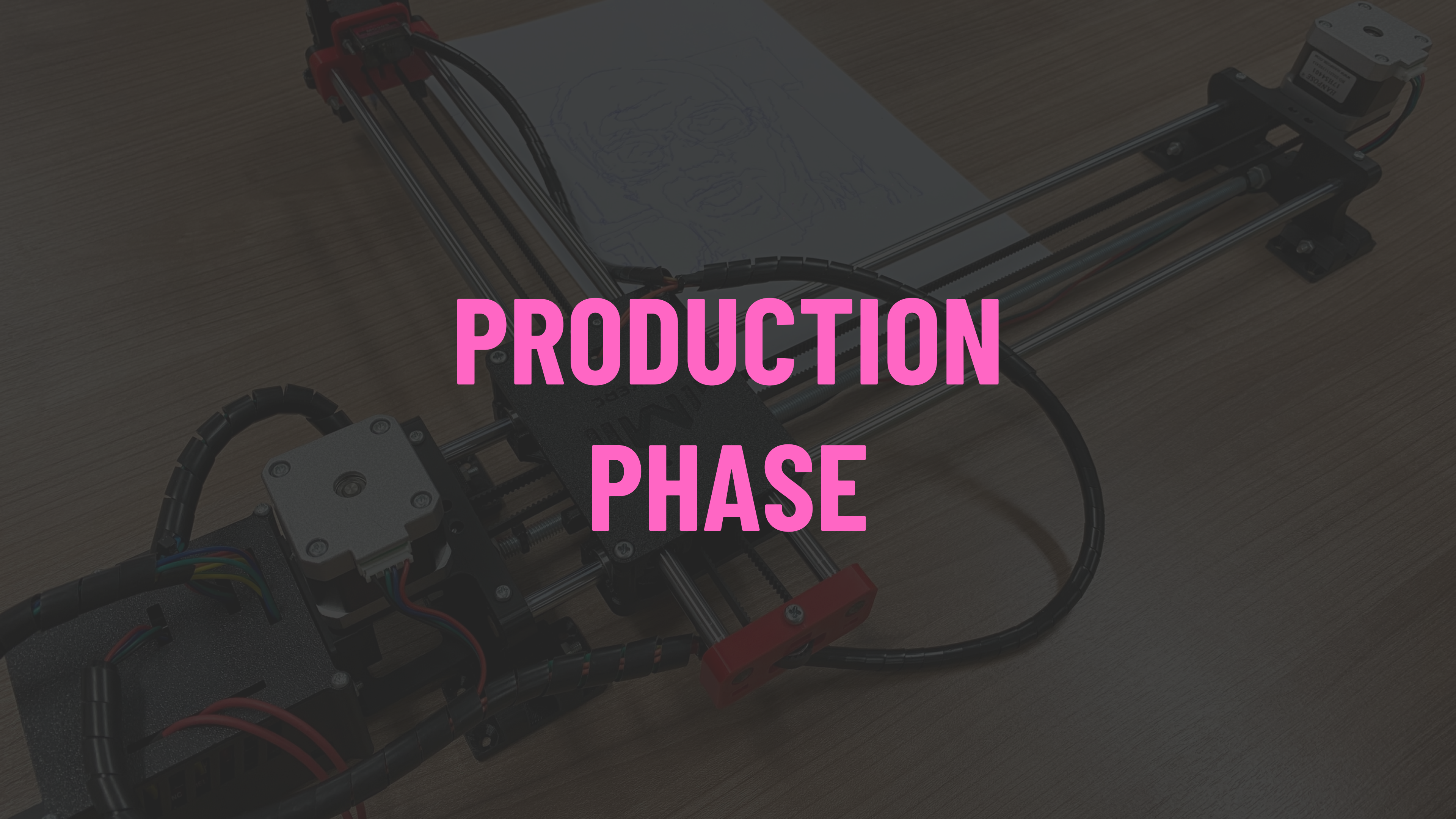
G1 - Linear Movement (X,Y)

G2 - Arc/Circular Movement (CW)

G3 - Arc/Circular Movement (CCW)

PROTOTYPING AND TESTING



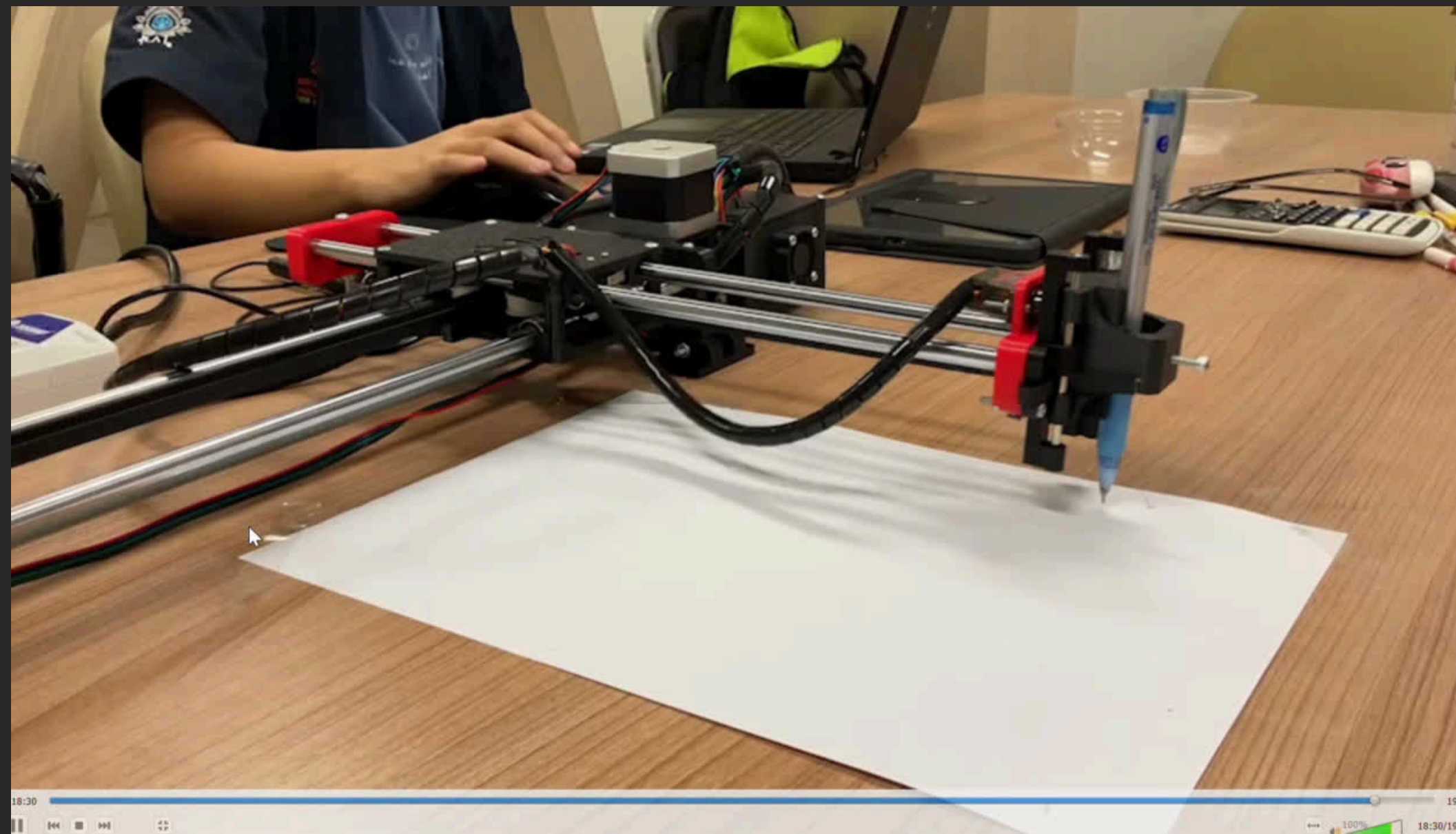
A 3D rendered image of a robotic arm assembly. The assembly consists of two servos, one at the top right and one at the bottom left, connected by a black cable. The servos are mounted on a black frame with silver-colored metal rods. A red servo horn is visible on the bottom left servo. The entire assembly is set against a dark, textured background. A semi-transparent pink rectangular box is overlaid in the center, containing the text "PRODUCTION PHASE" in a bold, white, sans-serif font.

PRODUCTION PHASE

TEAM MEMBERS



FINAL RESULT



REFERENCE AND USEFUL LINKS

- YouTube Link (<https://www.youtube.com/watch?v=NWOrIN7Ekjc>)
- DrawBot Original Uploader (<https://www.thingiverse.com/thing:2349232/files>)
- 3D Printers Kinematics(https://www.youtube.com/watch?v=_ramiM3KHYE&pp=ygUUdGVhY2hpbmcgdGVjaCBjb3JleHk%3D)